

Nishant Mishra

R&D Engineer –Thin film & Nanofabrication
Enschede, Netherlands
Nationality: Dutch
Languages: English (C1), Dutch (B1/B2)
[linkedin.com/in/NishantMishraEE](https://www.linkedin.com/in/NishantMishraEE)
nishant-mishra.com
nishant.mishra.nm@gmail.com



Professional Summary

R&D-focused equipment engineer with expertise in thin-film processing and characterization, tool maintenance, and workflow optimization. Experienced in training and supporting researchers, developing SOPs, conducting root-cause analysis, and automating workflows. Adept at ensuring tool reliability under diverse experimental conditions and enabling fast, safe, repeatable process development.

Technical & Core Skills

Nanofabrication & Cleanroom	Mask design; process-flow design; Mask and Maskless Lithography; RF Magnetron Sputtering; PECVD; Wet Etching; RIE; DRIE; Vapor Phase Etching
Workflow Optimization	Scripting and automation; cycle-time reduction; root-cause analysis
Characterization & Metrology	XRD; AFM; White-light interferometry; Electrical characterization (DC I–V, Hall, 4-point probe, EIS, chronoamperometry, DBLI ferroelectric testing)
Design & Development	Design of Experiments & Response surface mapping; Equipment commissioning, upgrades & maintenance; 3D modeling & printing
Safety & Operations	Pb safety; SOPs & training; sample traceability; inventory & supplier management; gas handling
Software	Python; MATLAB; C++ ; R

Professional Experience

University of Twente – XUV Optics Group

Research Engineer – Thin Film & Nanofabrication (Jul 2023 – Present)

- Trained & advised 8 researchers in sputtering, XRD, cleanroom processing, and mask design, accelerating onboarding.
- Automated XRD workflow: reduced operator time from 60 minutes to 10 minutes; improved signal-to-noise via program automation and optimization.
- Upgraded & automated sputtering system for remote, controlled operation (gas flows, shutters, pump-down, temperature ramps, magnetron power).
- Reduced cleanroom cycle time by 75% (from two half-days with overnight hold to half a day) by eliminating redundant steps.
- Authored & maintained 7 SOPs (sputtering, cleanroom processes, XRD, AFM, DBLI & I–V stress testing, equipment maintenance) to standardize training and reduce variability.
- Implemented sample transfer logbook ensuring traceability; achieved zero lost samples in 12 months.
- Managed inventory & supplier orders; accelerated commissioning of upgrades in collaboration with external suppliers and internal technical teams.
- Introduced lead-safety measures following safety review.
- Maintained 3D printer with zero downtime, 3D printed lab organizers to improve workflow efficiency.

ECsens (now OccamDx)

Research Intern – Nanostructure Devices (Jul 2020 – Dec 2020)

- Designed semiconductor masks and performed electrochemical experiments for biosensor development.

CARE, IIT-Delhi

Research Intern – Biomemristor Fabrication & Biosensors (Jun 2017 – Sept 2019)

- Developed biomemristor devices for neuromorphic applications.
- Conducted electrical characterization of bio-FETs and memristive devices; analyzed impedance spectroscopy data.

Twente Pathway College

Teacher – Physics | Module Coordinator – Computer Science | Module Coordinator – Engineering Project (Sep 2021 – Jun 2023)

- Coordinated and delivered modules for 100–150 students; supervised projects and assessments.
- Strengthened communication and stakeholder engagement across diverse international cohorts.

Education

M.Sc. Electrical Engineering – University of Twente, Enschede (2019–2021)

Specialization: Lab-on-Chip Systems | Research Honours Track

Master's thesis: Microvalve design for sub-millisecond gas transport to enable microsecond IR spectroscopy in photoelectrochemical CO₂ reduction; *Mechanism survey, device design & simulation, single-mask multi-depth cleanroom fabrication, heat-flow modeling, and control scheme.* [\[Link\]](#)

B.Tech. Electrical & Electronics Engineering – GGSIPU, Delhi (2014–2018)

Selected Certifications & Training

- [Lean Green Belt](#)
- [MOOC: Process Improvement and Optimization](#)
- [Basics and Design Principles for Ultra-Clean Vacuum](#)
- [Short Course: Reactive Magnetron Sputter Deposition](#)
- [Micro & Nano fabrication \(MEMS\)](#)
- [Software Carpentry Workshop: Git & Version Control](#)
- [Secure Handling of Gases – University of Twente](#)
- [Alarm and Firefighting – University of Twente \(in progress\)](#)

Publications

- Vancomycin functionalized WO₃ thin film-based impedance sensor for selective detection of Gram-positive bacteria. *Biosensors and Bioelectronics*, 2019. [\[Link\]](#)
- MoS₂/TiO₂ hybrid nanostructure-based FET for rapid detection of Gram-positive bacteria. *Advanced Materials Technologies*, 2019. [\[Link\]](#)